

**A
MINI PROJECT REPORT
ON**

**“Product Line Profitability & Margin Performance
Analysis for Nassau Candy Distributor”**

A Preliminary Internship Report Submitted To Unified Mentor



**Submitted By
Miss. Pranali Wakchaure**

2025-2026



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Dear Pranali Wakchaure

I'm pleased to offer you temporary employment as a **Data Science Intern** for a period of 6 months on behalf of Unified Mentor Pvt. Ltd. Starting from 10/01/2026 to 10/07/2026. If you agree to this proposal, your internship with the company will start right away. You'll have "temporary employment" status while you're an intern. All of the perks that permanent employees of the company receive are not available to you as a temporary employee.

By accepting this offer, you acknowledge that you understand participation in this program is not an offer of employment and successful completion of the program does not entitle you to an employment offer from Unified Mentor.

This letter supersedes all past conversations and agreements about your internship and is the final agreement between you and the Company. Only a written amendment that is endorsed by both of us may change the terms of this letter. We look forward to having you begin your career at Unified Mentor and wish you a successful internship.

Regards,

Drishti Madaan
HR Manager



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Authorized Signatory

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1. Acknowledgement:

I would like to express my sincere gratitude to my project guide Mr. Saiprasad Kagne and Faculty member for their valuable guidance & support throughout the completion of this project. Their continuous Suggestions and Encouragement helped me successfully complete this work. Additionally, I am thankful to all those who directly or indirectly contributed to the successful completion of this project.

2. Abstract:

This project presents a comprehensive data-driven analysis of product line profitability and margin performance for Nassau Candy Distributor. The primary objective is to identify high-performing and underperforming products using key financial metrics such as total sales, gross profit, profit per unit, and margin percentage. The dataset is preprocessed to handle missing values, ensure data consistency, and standardize product and division information. Various analytical techniques, including exploratory data analysis, division-wise performance evaluation, Pareto analysis, and cost versus sales diagnostics, are applied to extract meaningful insights. An interactive dashboard is developed using Streamlit to visualize important business indicators such as top profitable products, regional profit distribution, division performance, and profit contribution concentration. The results help in identifying profit-driving products, detecting cost inefficiencies, and supporting strategic decision-making. Overall, this project demonstrates how data science and business intelligence techniques can be effectively used to improve profitability analysis and optimize business performance.

[Keywords: Data Science, Business Intelligence, Profitability Analysis, Margin Analysis, Streamlit Dashboard, Pareto Analysis, Data Visualization, Financial Analytics.]

3. Introduction:

In today's competitive business environment, organizations must evaluate more than just sales performance to understand their overall financial health, as high sales volume does not always guarantee high profitability. For distribution companies such as Nassau Candy, certain products may generate significant revenue while contributing minimally to profit due to higher costs or lower margins, making it essential to analyze both revenue and cost components together. This project focuses on conducting a detailed analysis of product line profitability and margin performance using data science and business intelligence techniques.

Key financial metrics such as sales, cost, gross profit, and margin percentage are utilized to assess product and division-level performance. The dataset is carefully preprocessed to ensure accuracy, consistency, and reliability, followed by the application of exploratory data analysis and visualization techniques to uncover meaningful patterns and insights.

Furthermore, an interactive dashboard is developed using Streamlit to present the results in a structured and user-friendly manner, enabling dynamic exploration through filters and graphical representations. The objective of this study is to support data-driven decision-making by identifying high-performing products, detecting inefficiencies, and providing actionable insights that can help optimize product strategies and improve overall business profitability.

4. Problem Statement:

In distribution businesses such as Nassau Candy, relying solely on sales data can lead to incorrect evaluation of product performance, as high sales do not necessarily translate into high profitability. The organization lacks clear visibility into which products and divisions contribute significantly to profit and which ones suffer from low margins due to high costs or inefficient pricing. Additionally, there is limited insight into profit concentration, regional performance, and cost-to-sales relationships, making it difficult to identify inefficiencies and optimize decision-making. Therefore, there is a need for a data-driven analytical approach that evaluates key financial metrics and provides interactive visual insights to support better business decisions and improve overall profitability.

5. Objective:

The objective of this project is to analyze product line profitability and margin performance for Nassau Candy Distributor using data science techniques. The study focuses on identifying high-performing and low-performing products based on key financial factors such as sales, cost, gross profit, and margin percentage. It also aims to evaluate division-wise and region-wise performance, understand cost and sales relationships, and identify products that contribute the most to overall profit using Pareto analysis. Additionally, the project includes the development of an interactive Streamlit dashboard to visualize insights effectively and support better data-driven decision-making.

Following are some of objectives:

1. To study the relationship between cost and sales to identify inefficiencies.
2. To develop an interactive Streamlit dashboard for effective data visualization and decision-making.
3. To analyze region-wise profit distribution.
4. To develop an interactive Streamlit dashboard for effective data visualization and decision-making.

6. Literature Survey:

Previous studies in the field of business intelligence and data analytics highlight the importance of analyzing both sales and profitability to support effective decision-making. Researchers have shown that relying only on sales data can lead to misleading conclusions, as high revenue does not always indicate high profit. Various analytical approaches such as exploratory data analysis, margin analysis, and cost–revenue comparison have been widely used to evaluate product performance and identify inefficiencies.

Pareto analysis has also been commonly applied to determine the small percentage of products that contribute to the majority of profit, helping organizations focus on key profit drivers. In recent years, interactive data visualization tools and dashboards have gained significant importance, as they allow users to understand complex data through graphical representations and real-time insights. Technologies such as Python, along with libraries for data processing and visualization, and platforms like Streamlit, have made it easier to develop user-friendly dashboards for business analysis.

These studies emphasize that integrating data science techniques with visualization tools can significantly improve business performance by enabling data-driven strategies and better resource optimization.

7. Dataset Description:

The dataset used in this project contains detailed information about orders, products, customers, and financial performance for Nassau Candy Distributor. It includes multiple attributes such as Order ID, Order Date, Customer ID, Region, Division, Product Name, Sales, Units, Cost, and Gross Profit. These fields help in analyzing both operational and financial aspects of the business. Sales represent the total revenue generated from each order, while Cost indicates the production or procurement expense, and Gross Profit is calculated as the difference between Sales and Cost. The dataset also provides geographical information such as region and state, which enables region-wise performance analysis. Before analysis, the data is preprocessed to handle missing values, remove inconsistencies, and ensure accurate calculations. This structured dataset forms the foundation for performing profitability analysis, identifying key trends, and generating meaningful insights through visualization.

Dataset Preview Showing First 5 Records:

df.head()

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province	Postal Code	Division	Region	Product ID	Product Name	Sales	Units
0	1	US-2021-103800-CHO-MIL-31000	03-01-2024	30-06-2026	Standard Class	103800	United States	Houston	Texas	77095	Chocolate	Interior	CHO-MIL-31000	Wonka Bar - Milk Chocolate	6.50	2
1	2	US-2021-112326-CHO-TRI-54000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-TRI-54000	Wonka Bar - Triple Dazzle Caramel	7.50	2
2	3	US-2021-112326-CHO-NUT-13000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-NUT-13000	Wonka Bar - Nutty Crunch Surprise	10.47	3
3	4	US-2021-112326-CHO-SCR-58000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-SCR-58000	Wonka Bar - Scrumdiddlyumptious	10.80	3
4	5	US-2021-141817-CHO-	05-01-	05-07-	Standard Class	141817	United States	Philadelphia	Pennsylvania	19143	Chocolate	Atlantic	CHO-TRI-	Wonka Bar - Triple Dazzle Caramel	11.25	3

This figure presents a preview of the dataset, showing the first few records along with key attributes such as product name, sales, cost, and gross profit. It provides an overview of the data structure and helps in understanding the type of information available for further analysis.

8. Data Collection & Preprocessing:

The dataset used in this project is obtained from a structured source containing detailed information about product sales, cost, and profit for Nassau Candy Distributor. The data includes multiple attributes such as product name, division, region, sales, units, cost, and gross profit, which are essential for profitability analysis. Before performing analysis, the dataset undergoes preprocessing to ensure accuracy and consistency. Missing values are checked and handled appropriately, and invalid or zero-value records are removed to avoid misleading results. Data types are verified and corrected where necessary, and column names are standardized for better readability. Additional features such as margin percentage are calculated using existing fields to enhance analysis. These preprocessing steps help in improving data quality and ensure that the dataset is reliable for further analytical operations and visualization.

Dataset Preview Showing First 5 Records:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province	Postal Code	Division	Region	Product ID	Product Name	Sales	Units	
0	1	US-2021-103800-CHO-MIL-31000	03-01-2024	30-06-2026	Standard Class	103800	United States	Houston	Texas	77095	Chocolate	Interior	CHO-MIL-31000	Wonka Bar - Milk Chocolate	6.50	2
1	2	US-2021-112326-CHO-TRI-54000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-TRI-54000	Wonka Bar - Triple Dazzle Caramel	7.50	2
2	3	US-2021-112326-CHO-NUT-13000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-NUT-13000	Wonka Bar - Nutty Crunch Surprise	10.47	3
3	4	US-2021-112326-CHO-SCR-58000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-SCR-58000	Wonka Bar - Scrumdiddlyumptious	10.80	3
4	5	US-2021-141817-CHO-TRI-54000	05-01-2024	05-07-2026	Standard Class	141817	United States	Philadelphia	Pennsylvania	19143	Chocolate	Atlantic	CHO-TRI-54000	Wonka Bar - Triple Dazzle Caramel	11.25	3

The above figure shows a preview of the dataset, including key columns such as product name, sales, cost, and profit, which are used for further analysis.

Data Cleaning:

The dataset was checked for missing values using null value detection techniques, and it was observed that there were no missing or null values present in any column.

```
import pandas as pd

df = pd.read_csv("clean_nassau_candy_data.csv")

df.isnull().sum()

Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID 0
Country/Region 0
City        0
State/Province 0
Postal Code 0
Division    0
Region      0
Product ID  0
Product Name 0
Sales       0
Units       0
Gross Profit 0
Cost        0
Profit_per_Unit 0
Profit_Margin_% 0
dtype: int64
```

Data Preprocessing:

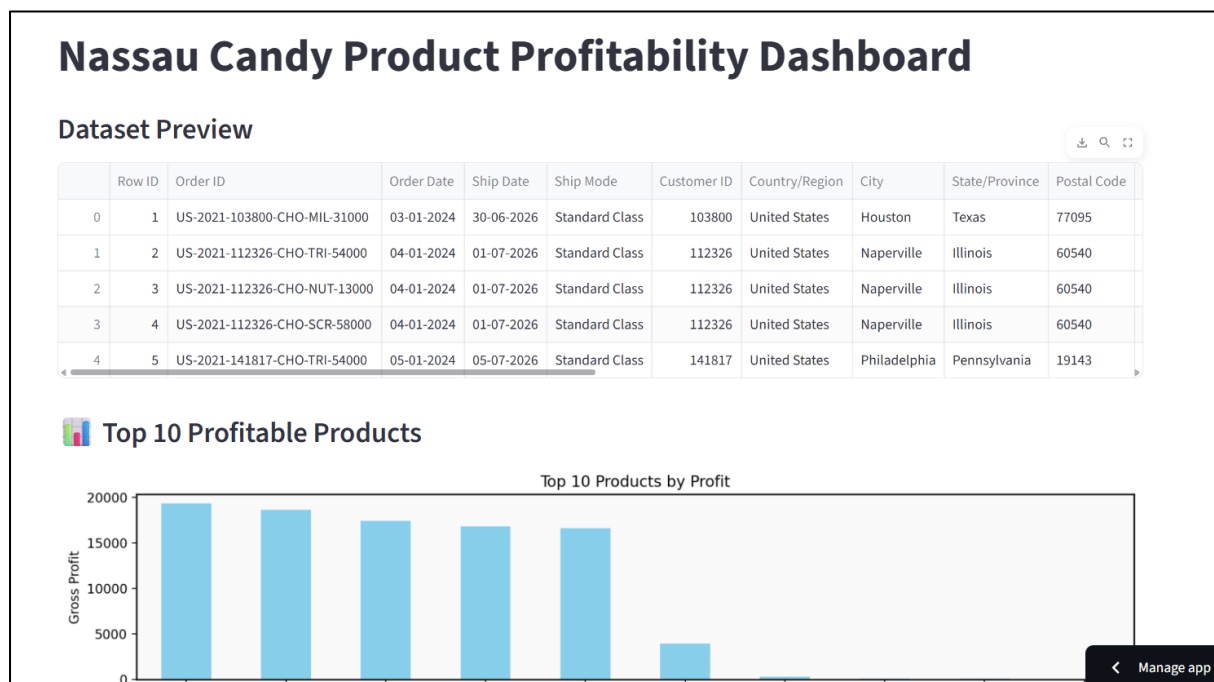
To ensure the accuracy and reliability of the analysis, the dataset underwent the following preprocessing steps:

1. Data Cleaning:
 - Removal of missing or null values to ensure data accuracy.
 - Elimination of duplicate and invalid records from the dataset.
2. Data Preprocessing:
 - Standardization of column names and data formats for consistency.
 - Conversion of data types (e.g., numerical and date fields) for proper analysis.
3. Data Filtering:
 - Filtering out records with zero or negative sales and profit values.
 - Selection of relevant columns required for profitability analysis.
4. Data Integration:
 - Combining data from different fields to create meaningful insights.
 - Deriving new attributes such as margin percentage using existing data.

9. Methodology:

The methodology followed in this project involves a structured approach to analyze product profitability and margin performance. Initially, the dataset is collected and preprocessed to ensure data quality by handling missing values, removing inconsistencies, and standardizing formats. After preprocessing, exploratory data analysis is performed to understand key patterns and relationships within the data. Important financial metrics such as gross profit, profit per unit, and margin percentage are calculated to evaluate product performance. Further analysis is carried out at both product and division levels to identify high-performing and underperforming segments. Pareto analysis is applied to determine the major contributors to overall profit, while cost versus sales analysis helps in identifying inefficiencies and pricing issues. Various data visualization techniques are used to represent insights clearly through graphs and charts. Finally, an interactive dashboard is developed using Streamlit to present the results in a user-friendly manner, enabling better understanding and data-driven decision-making.

Interactive Dashboard Overview:



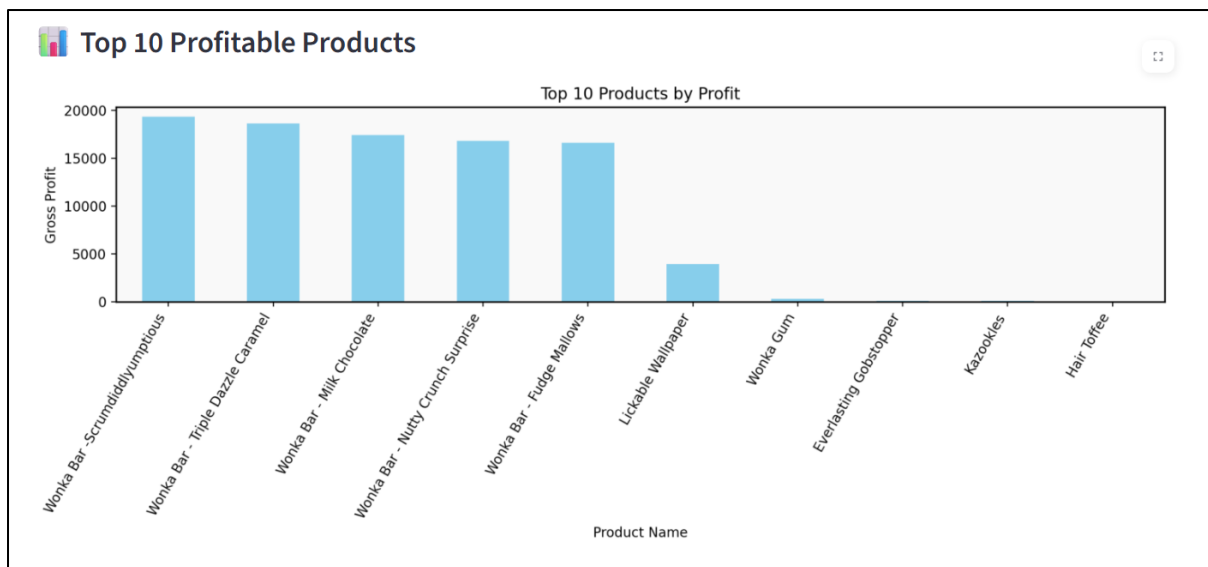
Data Visualization:

Data visualization is an essential part of this project, used to represent complex data in a clear and meaningful way. Various charts such as bar graphs and scatter plots are utilized to analyze product profitability, regional performance, division-wise comparison, and cost relationships. These visual representations help in identifying trends, patterns, and key insights effectively. The use of visualization makes the analysis more interactive and easy to understand, supporting better decision-making.

10. Dashboard Implementation & Tools-Technologies:

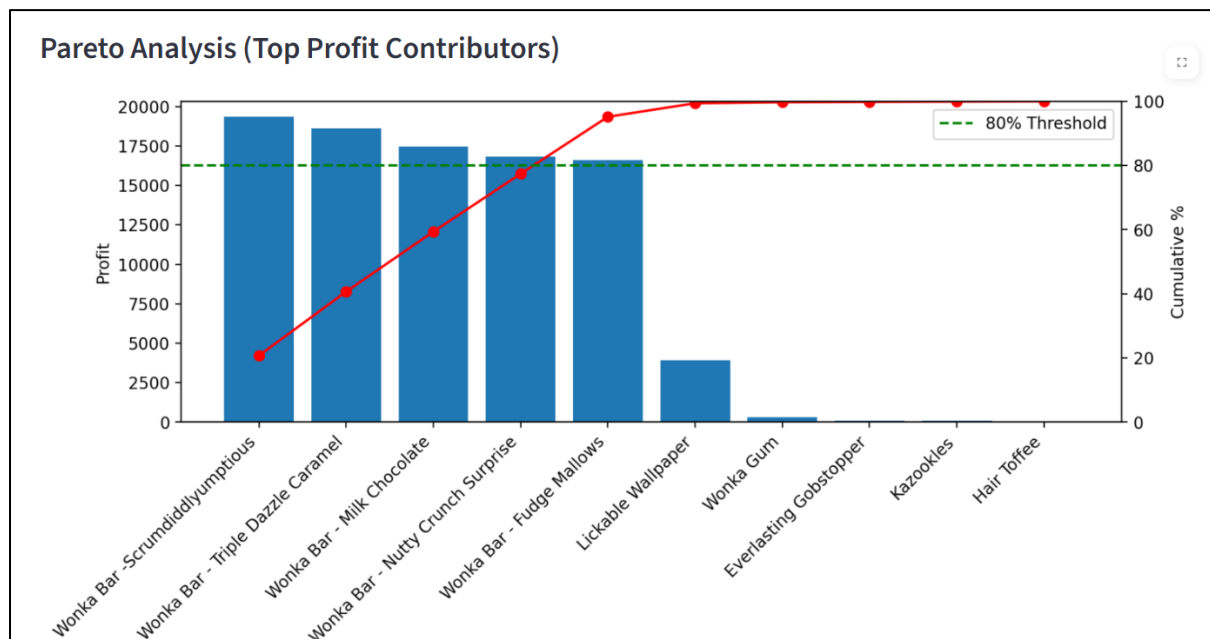
The dashboard for this project is developed using Streamlit to provide an interactive and user-friendly interface for analyzing product profitability and margin performance. The processed dataset is integrated into the application, and various visualizations are created using Python libraries to represent key insights effectively. The dashboard includes multiple analytical views such as top profitable products, region-wise profit distribution, division-level performance, Pareto analysis, and cost versus sales relationships. Interactive features like filters and dynamic selections are incorporated to allow users to explore data based on specific criteria such as division and product. The visualizations are designed to present complex data in a simple and understandable format through bar charts and scatter plots. This dashboard enables users to quickly identify high-performing products, detect inefficiencies, and gain actionable insights, thereby supporting data-driven decision-making and improving overall business performance.

Top 10 Profitable Products Showing Highest Gross Profit Contribution:



This graph highlights the top 10 products that generate the highest gross profit. It helps in identifying the key products contributing most to overall profitability. These products can be considered as major revenue drivers and should be focused on for business growth and strategic decision-making.

Pareto Analysis Showing Profit Contribution:



Profit Distribution Across Different Regions:



Tools & Technologies:

The project is implemented using various data science and visualization tools to ensure effective analysis and presentation of results. Python is used as the primary programming language due to its simplicity and strong support for data analysis. Libraries such as Pandas are utilized for data manipulation and preprocessing, while Matplotlib is used for creating graphical visualizations. The dataset is processed and analyzed using Jupyter Notebook for exploratory analysis and development. Additionally, Streamlit is used to build an interactive web-based dashboard that allows users to visualize insights dynamically.

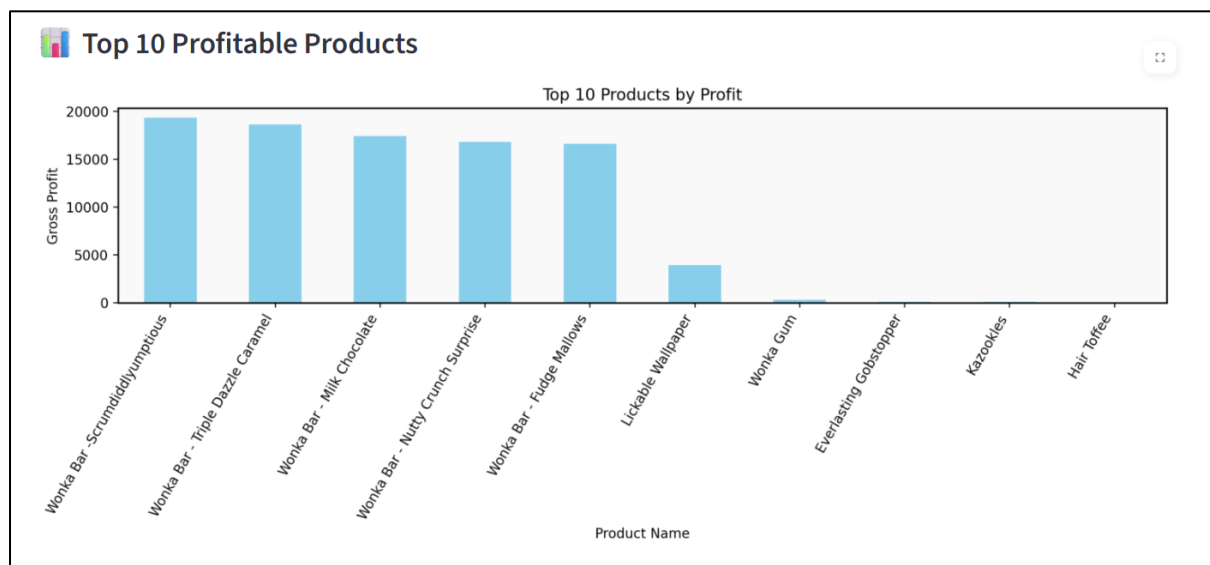
11. Result & Analysis:

The analysis of the dataset provides several important insights into product profitability and overall business performance. The Top 10 Profitable Products graph highlights the products that contribute the highest gross profit, helping in identifying key revenue drivers. The region-wise profit distribution shows how profit varies across different regions, indicating areas with strong and weak financial performance. Division-level analysis reveals differences in profit and margin percentage among various product categories, allowing the identification of high-performing and low-performing divisions. The Pareto analysis demonstrates that a small number of products contribute to a large portion of total profit, supporting the 80/20 principle and helping in focusing on major profit contributors.

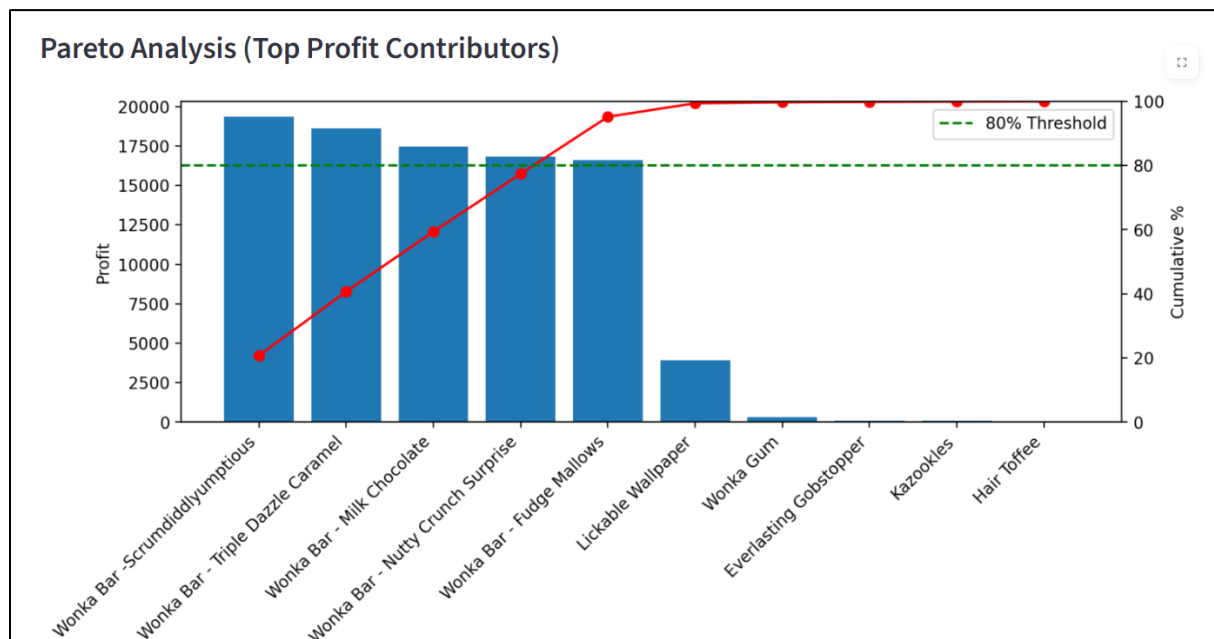
Additionally, the cost versus sales scatter plot helps in identifying products with high costs but relatively low sales, indicating potential inefficiencies in pricing or cost management. The units versus profit relationship provides further insight into how sales volume impacts profitability. Overall, these analyses help in understanding key trends, identifying profitable segments, and detecting areas that require improvement, thereby supporting better data-driven decision-making.

Graphical Analysis and Visualization:

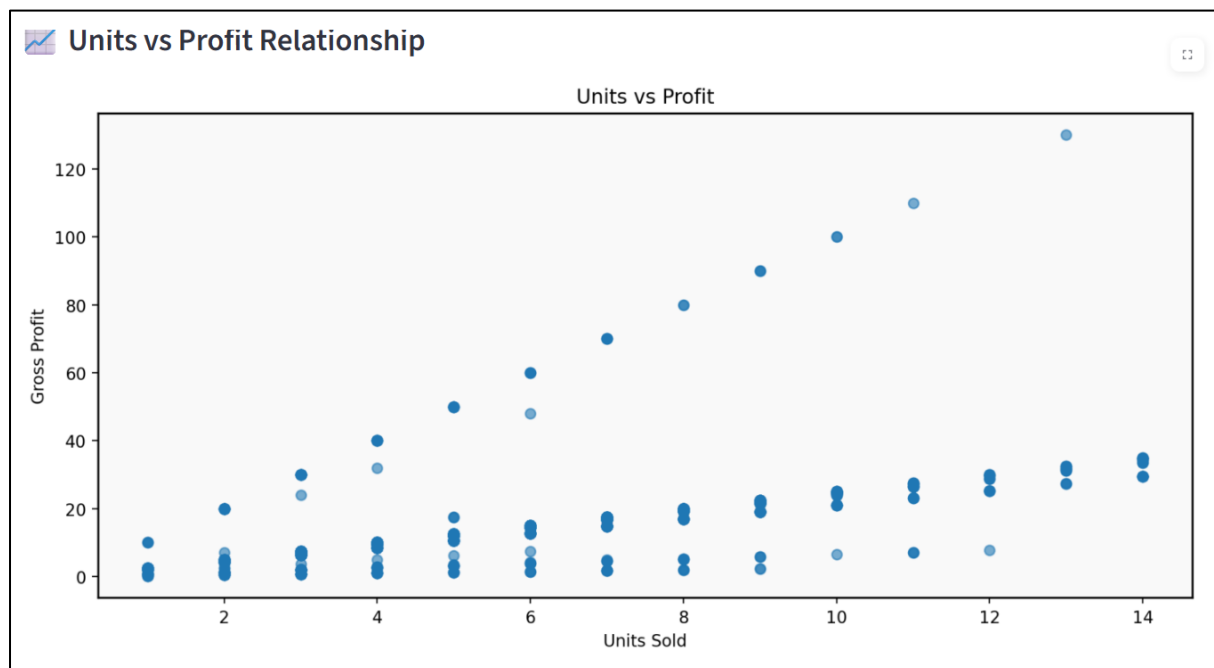
1. Top 10 Profitable Products Based on Gross Profit



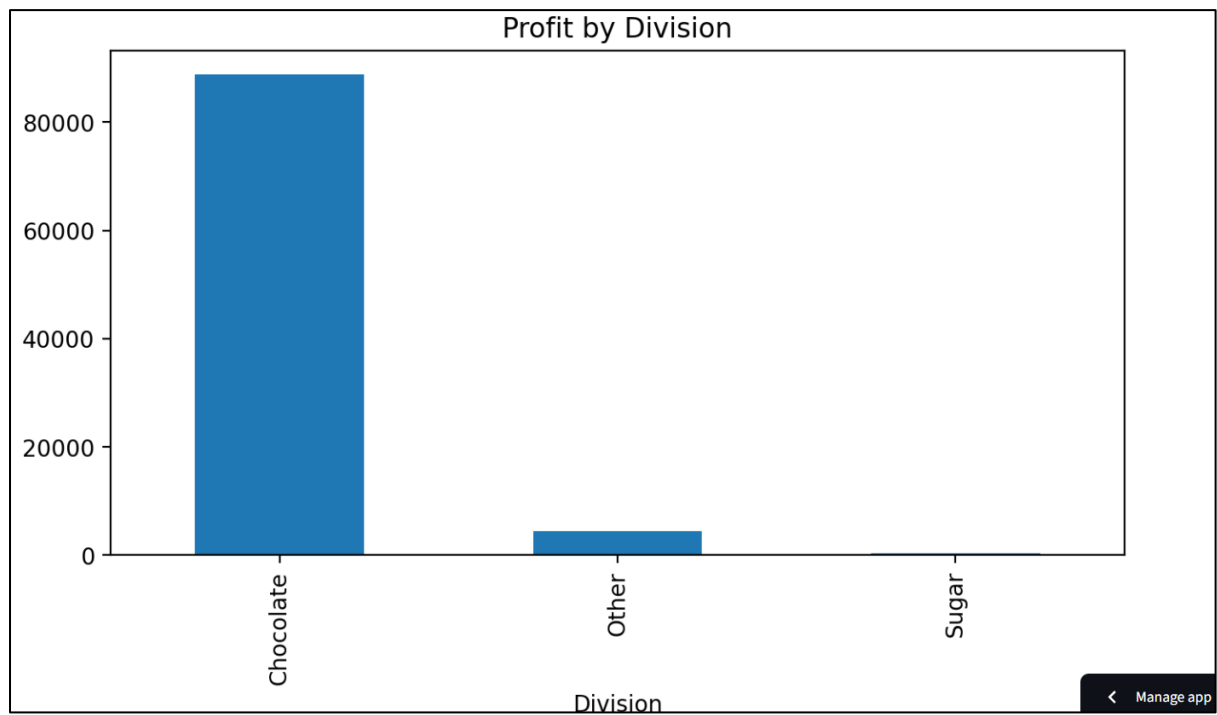
2. Pareto Analysis (Top Profit Contributors)



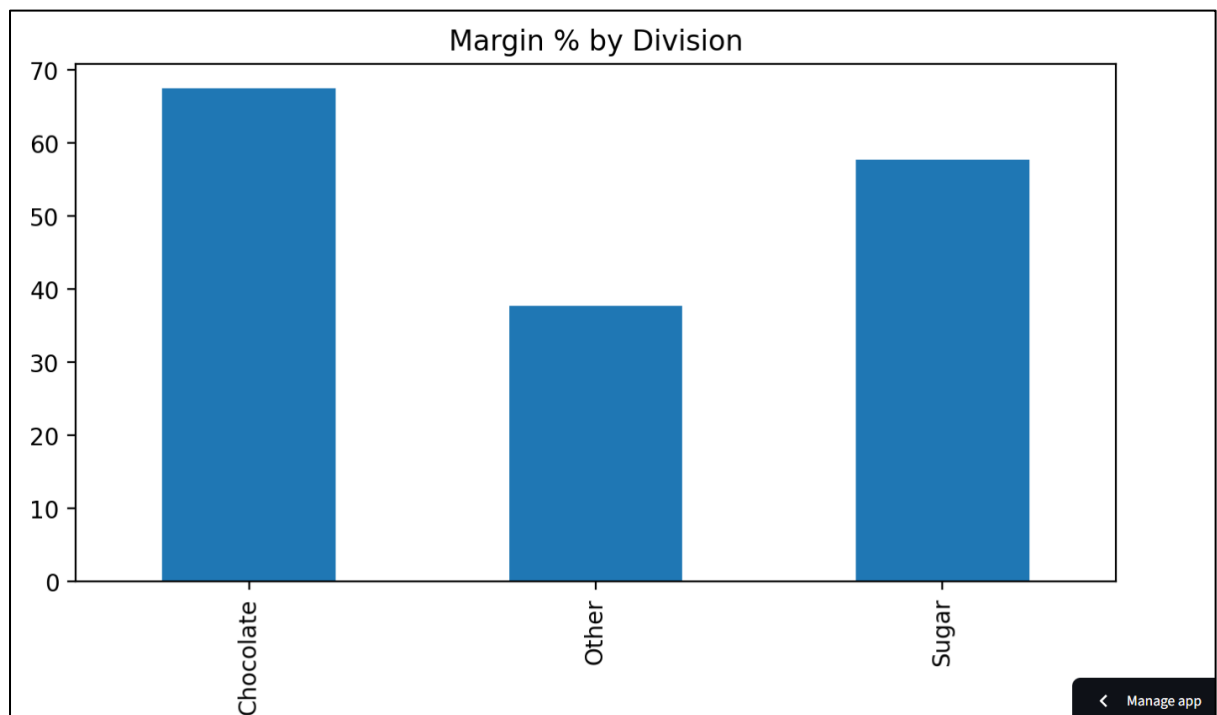
3. Units VS Profit Relationship



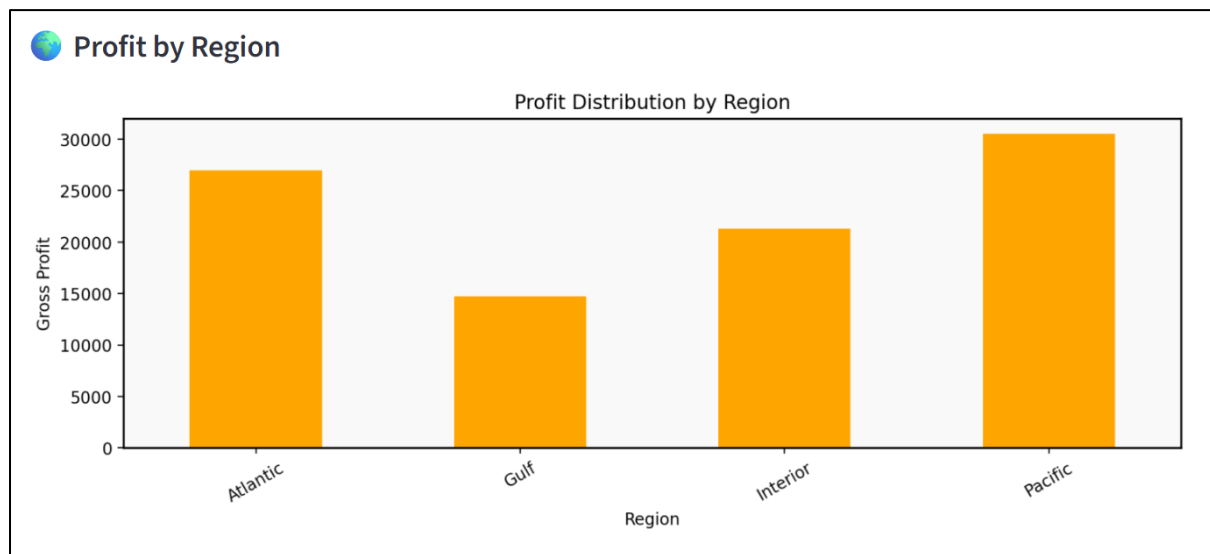
4. Division Performance



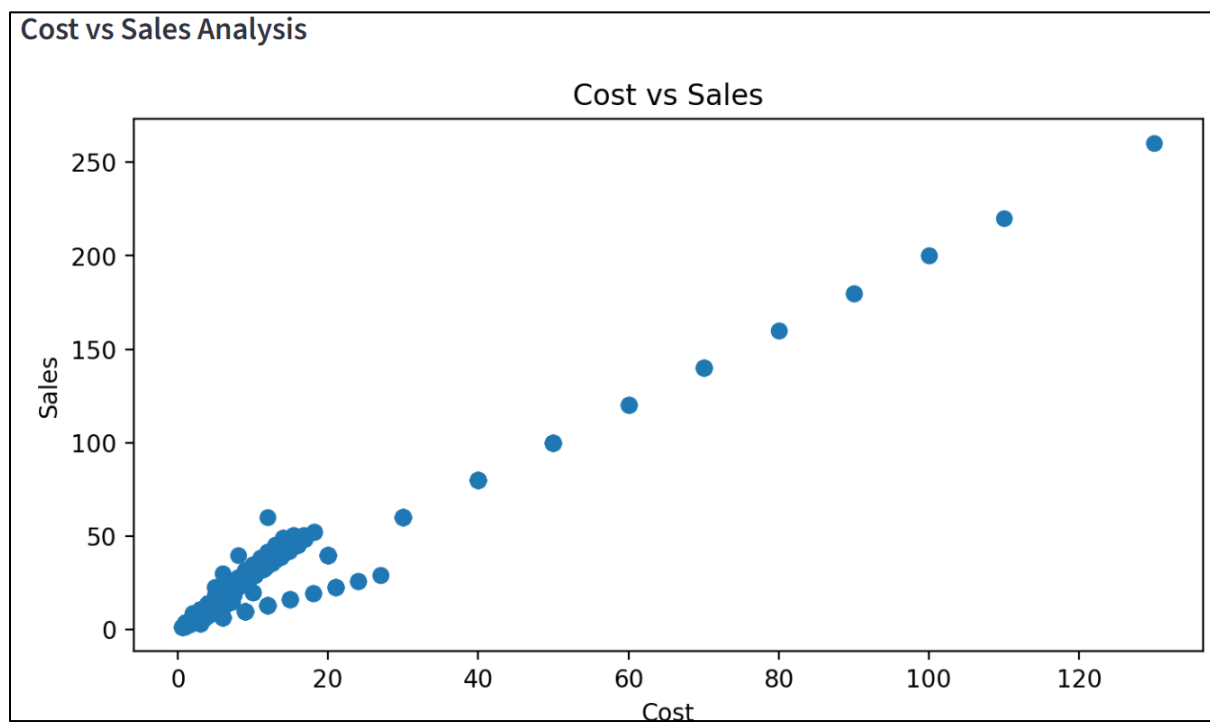
5. Margin % By Division



6. Profit Distribution Across Different Regions



7. Cost VS Sales Analysis



12. Conclusion & Recommendations:

Conclusion:

The analysis of Nassau Candy Distributor's dataset provides meaningful insights into product profitability and overall business performance. The results show that only a limited number of products contribute to a major portion of total profit, while some high-selling products generate relatively lower margins. Division-wise and region-wise analysis highlights variations in performance across different segments of the business. The cost versus sales relationship further helps in identifying inefficiencies in pricing and cost structure. Overall, the study demonstrates the importance of analyzing both revenue and cost factors to understand true business performance and supports the use of data-driven approaches for effective decision-making.

Recommendations:

Based on the analysis, it is recommended that the company should focus on promoting high-margin and high-profit products to maximize overall profitability. Pricing strategies for low-margin products should be reviewed and optimized, and efforts should be made to reduce production or procurement costs where possible. Underperforming products should be carefully evaluated for improvement or potential discontinuation. Additionally, the company should leverage interactive dashboards for continuous monitoring and better decision-making. Implementing these strategies can help improve financial performance, optimize product portfolios, and enhance overall business efficiency.

13. Future Scope:

The current project can be further enhanced by incorporating advanced data analytics and machine learning techniques to improve prediction and decision-making capabilities. Future work may include the use of predictive models to forecast sales and profitability trends, enabling proactive business strategies. Integration of real-time data can also be implemented to provide up-to-date insights and continuous monitoring of performance. Additionally, more advanced interactive features and filters can be added to the dashboard to improve user experience and flexibility. Expanding the analysis to include customer behavior, seasonal trends, and supply chain factors can further enhance the depth of insights. Overall, these improvements can make the system more robust, scalable, and valuable for strategic business planning.

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